Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/19/2006
PI Name:	Vuletich, David Ph.D.		
Project Title:	Using cyanobacterial hemoglobins to sustain human life on Mars		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
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Human Research Program Elements:	None		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal Text:			
Task Description:	Human exploration of and inhabitation of other terrestrial planets is limited by the need for continuous supplies from Earth. The establishment of a permanent habit on Mars would be greatly simplified if the natural resources of the planet could be harvested, thereby eliminating the need for constant supply or perfect recycling. U.S. Patent # 4,343,715 describes a system for sequestering O2 from seawater using a polymer matrix made of foam or gel which contains immobilized hemoglobin molecules. This idea could be used to design an O2 extraction device for use on Mars. The most important component of this device would be the O2 extraction molecule. The hemoglobin from the cyanobacterium Synechococcus sp. PCC 7002 (S7002 Hb) has certain desirable properties for this purpose. These properties include a high O2 affinity for extraction from the low partial pressure of the Martian atmosphere and a high stability to withstand possible stresses from the extraction device and Martian environment. The current study will examine methods for improving S7002 Hb stability along with determining what reducing agents may be used to facilitate bound O2 release. Also, an examination of the structural and electronic changes that occur upon small		

molecule binding will be conducted. In the second year of funding, the focus of this project include included studies to modulate the affinity of the protein for oxygen and carbon monoxide. In its current state, S7002 trHb most likely binds carbon monoxide with a higher affinity than oxygen. This problem will have to be overcome because the Martian atmosphere has about half as much CO as O2, which is not a large enough threshold. In addition to the CO discrimination, examination of factors that contribute to the protein stability was extended. These goals were accomplished through detailed phylogenetic analysis, characterization of novel proteins, and hydrogen-deuterium exchange NMR to explore the strength of hydrogen bonding interactions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Completion of this project will provide future Mars astronauts a method to sequester oxygen from the natural environment that will not require air purification and recycling machinery.
Also, the detailed phylogenetic analysis of this protein family gives insight into the history of the hemoglobin gene. Understanding the changes that occur throughout evolution can impact studies of hemoglobin related disease and the development of blood substitutes.

Task Progress:

The research goals during the second year of this project involved examining the cyanobacterial hemoglobin with respect to various other members of the protein family through specific aims 1 and 2, and to gain an understanding of the added stability of the heme-protein cross-link through specific aim 3.
Specific aim 1: Phylogenetic analysis of the truncated globin family

An original phylogenetic study on the truncated globin family showed that these proteins were split into three distinct phylogenetic subgroups (N, O, and P) (1). Since the time of this study, a few members of two of the three subgroups have been biophysically and structurally characterized. Of these proteins, the truncated globin from Mycobacterium tuberculosis of the O subgroup was found to have a higher oxygen affinity than carbon dioxide affinity (2). If this characteristic could be implanted into S7002, both the advantage of added stability due to the cross-link and the CO discrimination could be instituted into one molecule. In order to assess what changes would need to be made to the S7002 primary structure, detailed sequence alignments are needed to find the corresponding positions between the proteins. The trHbs from all three subgroups have potential to reveal unique ways of stabilizing and discriminating ligands along with novel ways to improve protein stability or modulate redox potential. Discoveries are being made at a rapid rate in the trHb field and it would be beneficial to take advantage of any desirable characteristics as quickly as possible. In order to do this, a detailed phylogenetic analysis of all the current trHb sequences needs to be completed. Accurate sequence alignments will allow for direct comparison between positions in all the proteins so decisions about what residues to vary will be easy to make when a particular protein trait is desired. The phylogenetic analysis will also reveal which residues are highly conserved and therefore presumed to be important for ligand binding and structural stabilization. This study will serve as a basis for all future decisions involving variation of the S7002 protein to obtain a desired trait of another trHb.

Progress:

The first aim, a phylogenetic analysis of the 2-on-2 globin family, found that these proteins split into three groups of orthologs and that each of the groups was paralogous to one another. This analysis lead to the conclusion that each group should be examined separately to look for conserved features that potentially detail some functional aspect. This was completed, and various conservation patterns were discovered that likely determine the ligand binding characteristics within each group. Some of these characteristics involved ligand stabilization and discrimination, both of which are important to the design of an oxygen scavenging system for the Martian atmosphere. It seems very possible that aspects of any of the proteins in this family could be combined unto the S7002 Hb scaffold, which would provide a very stable oxygen binder that discriminates between oxygen and carbon monoxide. In addition, much was learned about the history of the 2-on-2 globin history, which serves to aid the overall history of the globin gene.

Specific Aim 2: Characterization of a trHb P Globin

Of the trHb family, only the N and O globins have been studied to any extent. These trHbs all have unique ligand binding characteristics and other properties such as cooperative oxygen binding that could be useful in an oxygen extraction system. Therefore, it seems necessary to examine the P globin subgroup as initial inspection of primary structure indicates a unique and highly regulated heme pocket. Thus, the goal of the second specific aim will be to clone and express a P globin gene from Helicobacter hepaticus and complete the initial NMR and EPR studies of the heme pocket. Ligand binding studies will also be completed. Any desirable characteristics could then be identified on a residue basis, and the phylogenetic analysis will be able to determine what changes would need to be made in S7002 to obtain those characteristics. By having at least one member of each subgroup well studied, all the positive attributes for successful Martian oxygen extraction can be exploited in the most structurally sound framework.

Progress:

Specific aim 2 was not completed due to problems with the preparation of the protein. These problems are being addressed and solved.

Specific Aim 3: Hydrogen-Deuterium Exchange in S7002 It is known that the formation of the heme-protein covalent-link in S7002 increases the melting temperature of the protein by about 20 °C (3). This is an extremely desirable feature for this system because added stability will help in the harsh Martian climate to prevent unfolding of the protein and heme loss. Understanding the factors that increase protein stability is important if optimal stability is to be obtained. One characterization method consists in monitoring exchangeable hydrogens, i.e., those that can be replaced by solvent hydrogens on a practical time scale. The amide hydrogens of the protein backbone are used as probes in experiments exposing the protein to heavy (2H) water. In its most efficient form, the study requires 15N labeled protein so both the amide proton and nitrogen can be examined using NMR spectroscopy. The protein is first put in a solution of 1H2O so all the amides are the regular NMR active nuclei. Then the sample is dried and resuspended in 2H2O. As the amide 1H’s are replaced by 2H’s, the corresponding 1H NMR signals decrease and eventually reach the isotopic composition of the solvent. It is generally observed that the hydrogen exchange rate is retarded in relation to the

	stability of hydrogen bonds and ultimately the structure of the protein. A comparison of exchange rates in the non-cross-linked form and the cross-linked form of the protein will provide insight into the region of the structure that are affected by the cross-link, and therefore reveal where additional changes should be made to increase the stability further. Progress: The hydrogen-deuterium exchange experiments showed that the stabilization of S7002 through cross-linking was not a general feature of every amide in the protein. Particular regions were either stabilized or destabilized. Of the regions that were stabilized, the most important was the proximal side of the heme. In addition, the distal, or ligand binding side, was destabilized. Due to the endogenously hexacoordinate nature of the protein, this result would imply more rapid ligand binding in the non-crosslinked protein if hexacoordination is a factor in the exogenous binding. This result was reported for an ortholog of S7002 Hb that is also hexacoordinate and can crosslink the heme. The nonuniform nature of stabilized also implied that the can in stability from crosslinking is most likely an entropic effect. This means that a crosslink engineered into any heme protein could result in increased stability. (1) Wittenberg, J. B., Bolognesi, M., Wittenberg, B. A., and Guertin, M. (2002) Truncated hemoglobins: a new family of hemoglobins widely distributed in bacteria, unicellular eukaryotes, and plants. J Biol Chem 277, 871-4. (2) Ouellet, H., Juszczak, L., Dantsker, D., Samuni, U., Ouellet, Y. H., Savard, P. Y., Wittenberg, J. B., Wittenberg, B. A., Friedman, J. M., and Guertin, M. (2003) Reactions of Mycobacterium tuberculosis truncated hemoglobin O with ligands reveal a novel ligand-inclusive hydrogen bond network. Biochemistry 42, 5764-74. (3) Vu, B. C., Vuletich, D. A., Kuriakose, S. A., Falzone, C. J., and Lecomte, J. T. (2004) Characterization of the heme-histidine cross-link in cyanobacterial hemoglobins from Synechocystis sp. PCC 6803 and Synechococcus sp. PCC 7002. J Biol Inorg Chem 9, 183-94.
Bibliography Type:	Description: (Last Updated: 09/26/2006)
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